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Early Diagnosis of Cardiac Diseases using ECG Images and CNN-2D

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Abstract

Heart disease is becoming the biggest cause of mortality worldwide. Its early detection can considerably lower the risk of mortality and help to promote its successful treatment. However, this early detection necessitates regular monitoring of a wide range of clinical and lifestyle factors. This is why a growing number of studies are being conducted to automate the forecasting of cardiac diseases, beginning with an examination of ECG images, which is the first diagnostic test performed on patients and also the most simple and economical to conduct. This study investigates the use of three groups of ECG images acquired from three separate sets of cardiac patients, with different heart-related illnesses, and a set of healthy controls to predict heart disease using deep learning classifiers. The evaluation is carried out on a real-life dataset, and the results highlight really interesting findings.

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1. Introduction

Cardiovascular diseases are some of the principal causes of morbidity, disability, and mortality. For this reason, predicting the onset of the disease timely is essential. Most diseases are diagnosed through the examination of the electrocardiogram (ECG), the graphic reproduction of the electrical activity of the heart during its operation. The electrocardiogram is a diagnostic procedure that involves the use of one or more sensor devices for recording and visually reporting the heart rhythm as well as its electrical activity. In order to record an ECG, 12 electrodes are usually positioned on the surface of the body to detect voltage changes that are, in turn, reported in an ECG image,

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where it is possible to visualize the trend of the obtained signal and quantify it (Figure 1). In particular, an ECG image records the bio-electrical activity of the cardiac system, and when heart disease occurs, some or all of the signals provided by the electrodes differ from their normal state [1, 2]. ECG is preferred to other diagnostic tests such as cardiac magnetic resonance, diagnostic coronary angiography, or myocardial scintigraphy, for multiple reasons: the examination is performed in a few minutes, the delivery of the report after a maximum of half an hour, it can be performed at any age, and it is a non-invasive exam. For all these reasons, ECGs play a fundamental role in scientific research and have been frequently studied. Heart disease can be identified by examining the ECG pattern; as a matter of fact, an “arrhythmia” is a set of irregular cardiac impulses.

The study proposed in this paper uses a new approach considering 12-lead-based ECG images, instead of the bio-electrical signals themselves, in order to accelerate the time to train and simplify the pre-processing phase. We performed different types of analysis: first, we use, for the first time, as far as we know, the whole ECG image by performing both binary classification (sick/healthy patient) and the multiclassification of the specific type of heart disease; we then analyzed the images divided into three bands, following the distribution present on the trace, in order to understand if there is a set of electrodes that is able to better capture the differences between the different arrhythmias. In order to automate the detection of cardiac disorders, we fed a 2D Convolutional Neural Network (CNN-2D) with the aforementioned input, achieving appreciable and interesting results.

The novel and relevant contributions of this paper are the following:

- Our approach employs scanned ECG images instead of bio-electrical signals, in order to simplify and speed up the prediction system, which produces good performance with lesser pre-processing, memory, and training time than other traditional approaches.
- Our analysis strategy diagnoses the onset of a cardiopathy early and, at the same time, is able to characterize the disease by recognizing distinct types of arrhythmias.

The rest of the article is organized into further six sections, as follows: Section 2 offers an overview of the current literature on the issue; Section 3 introduces and describes the employed deep neural network models. Section 4 describes the datasets (Subsection 4.1), all the characteristics of the experiments conducted in this work, and the methodology of validation of the results (Subsection 4.2); while Section 5 highlights all the results achieved as a consequence of this study. Finally, all possible threats to the validity of the study and its results, as well as the conclusions reached by our work and future research possibilities, are illustrated in Sections 6 and 7, respectively.

2. Related Work

In the relevant literature, many studies are related to the automatic identification of heart disease through ECG signals [3, 4]. The characteristics derived from the cardiac beat and the selected classification algorithms are the two most crucial components in a heart disease prediction system [5]. Many parameters have been retrieved from the ECG to categorize heartbeats: morphological traits, wavelet transformation, Fourier transform, and statistical parameters [5, 6, 7, 8, 9, 10].

With the aim of creating a bridge between human beings and technology, in the last decades, the research community has concentrated on Artificial Intelligence (AI) by working on digital image processing, sound recording, clinical data, and many types of data to diagnose different cardiac pathologies [8, 11, 12]. As regards ECG images, according to the current literature, they are usually reconfigured as signals on which to carry out the analyses and are not used as images properly. These signals are analyzed, in most works, through Deep Learning (DL) techniques. DL handles dimensional ECG data as a time series categorization and it is the most used AI-based technique to detect cardiopathies; furthermore, DL has achieved growth potential with excellent results in medical image detection and classification tasks [12]. In particular, Convolution Neural Network (CNN) is regarded as a cutting-edge method for detecting and classifying cardiac signals, and it has been explored in several versions such as 1-dimensional (1-D), 2-dimensional (2-D), or a mix of both [13, 14]. Noman et al. [14] developed a framework for direct feature learning from raw cardiac signals based on TF-ECNN, a combination of CNN-1D and CNN-2D using 2-dimensional time-frequency (TF) feature maps. In order to evaluate the performance of their architecture, they also classified the heartbeat with more

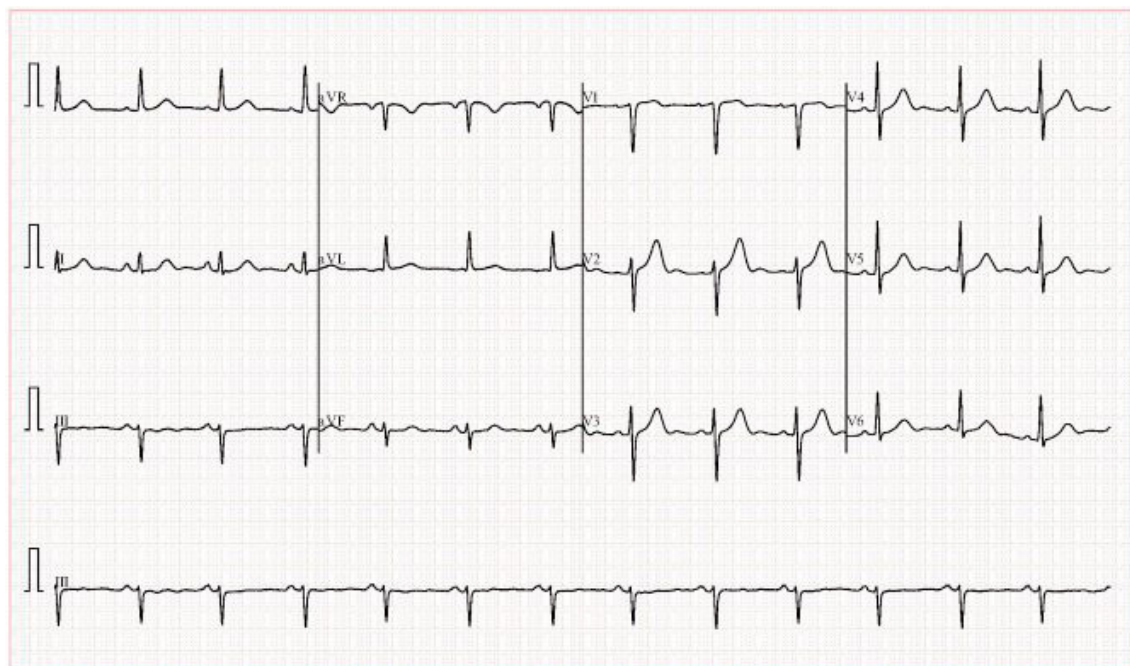


Fig. 1. Example of an ECG image.

traditional classifiers such as a Support Vector Machine (SVM), an ensemble model of Decision Trees (DTs), and Hidden Markov models (HMMs). Their study shows that TF-ECNN is the best classifier. Huang et al. [16] converted five types of cardiac beats into time-frequency spectrograms for training a deep CNN-2D with the aim of distinguishing various arrhythmia types. In order to evaluate their results with CNN-2D, they also compared the proposed model with a traditional CNN-1D model. The comparison showed that the CNN-2D model is more accurate than the CNN-1D one. Lu et al. [17] and Ji et al. [18] both converted 1D-heartbeat signals into 2D pictures to classify the ECGs, the former using a Random Forest classifier, while the latter by means of CNNs. Moreover, Ferretti et al. [25] proposed ECG classification of different arrhythmia types based on CNN-1D. They compared four different configurations of the proposed neural network to find the best model. In particular, the configuration with the highest accuracy had five different layers, two of which were fully connected.

The studies reported in this section exhibit that ECG is often examined by converting the images into signals and not analyzing the image itself. Differently from the reviewed literature, the study we propose defines a novel approach based on the 12-lead-based ECG images instead of the electrical signals, this is particularly innovative in our humble opinion. Furthermore, the approach is based on a CNN model with 2 dimensions, the most used network to analyse images, it is based on scanned images of the ECGs, and it is also capable of predicting, in an automated way, specific cardiac pathologies as reported in the relevant literature.

3. Background

3.1. Heart disease

A healthy heart supplies the right amount of blood for the entire body to function correctly. The different parts, of which it is made up, work in a coordinated way in each instant of the life of the human being. When there are heart lesions or diseases, the organs would no longer receive the oxygen they need to work properly. Heart disease, a term for all heart pathologies, can interfere with heart pump capabilities. Moreover, its work may be complicated by ailments of structural nature, such as damaged blood vessels, or by functional disorders, such as problems related to the electrical system of the heart, e.g., arrhythmias. Heart disease is a very varied category of pathologies that can

be classified into: *Congenital heart disease*, a disease inherent in defects present at birth, often due to genetic factors; *Acquired heart disease*, an illness that arises following the presence of other pathologies due to incorrect behaviors.

In particular, the cardiac diseases analyzed in this work are the following two:

- *Myocardial Infarction* (MI)[20]: the term refers to a myocardial injury that is accompanied by clinical evidence of acute myocardial ischemia and the observation of an increase or decrease in cTn values¹. In addition, one or more of the following symptoms must occur: symptoms of myocardial ischemia, new ischemic electrocardiogram changes, pathological Q-wave development, imaging evidence of new loss of viable myocardium, new regional wall motion abnormality in a model consistent with an ischemic aetiology, and identification of a coronary thrombus by angiography or autopsy.
- *Abnormal HeartBeat* (HB)[21]: pulse, another name for heartbeat rate, is the number of times a person's heart beats in a minute. The typical heart rate changes depending on the individual. Normally, an adult's heart should beat between 60 and 100 times per minute. When the heart beats too slowly, too fast, or irregularly, there is an arrhythmia. There are different types of cardiac arrhythmias: bradycardia when the heartbeat rate drops below 60 beats per minute and therefore beats more slowly; tachycardia (atrial/ventricular) if the beat rate is greater than 100 beats per minute, therefore it goes faster; and beat arrhythmias, in which the rhythm of the beat is altered, not the speed, as happens in the case of extra-systole, i.e., the presence of single or repetitive extra beats.

3.2. Convolutional Neural Network

Deep learning has achieved really good performance over the past few years on a range of issues, including speech recognition, image recognition, and natural language processing [22, 23]. In particular, Convolutional Neural Networks (CNNs) are among the varieties of deep neural networks that have been the subject of the most research. A CNN is a feed-forward neural network used to analyze visual pictures by processing data in a grid-like architecture. It is composed of numerous stages or layers, each one focused on a distinct functional aspect. Unlike the neurons of a typical neural network, the neurons of a CNN are organized in three dimensions: width, height, and depth. In particular, the neurons in a layer will not all be entirely linked to the neurons in the layer preceding it; for instance, if the input picture has a 32x32x3 dimension (width, height, and depth, respectively), the final output layer will be 1x1x10 in size since the CNN will eventually condense the whole image into a single vector of class scores that are ordered along the depth axis.

A CNN is typically made up of a series of ordered blocks, each one appointed to a layer of the network that is in charge of changing the volume of activations into another layer, using a differentiable function.

The main layers of a usual CNN are the following:

- *Input layer*: it is the collection of data in the shape of numbers to be evaluated.
- *Convolutional layers (Conv)*: these layers use numerous filters used to extract the essential characteristics of the input data.
- *ReLU layers (Rectified Linear Units)*: they bring non-linearity into the network by calculating linear operations using the scalar product between the filter and the receptive field during the various convolutional stages. The result is a modified feature map that has been linearized.
- *The Pool layers*: they control the lowering of the convoluted feature's spatial size (height and breadth). Through dimensionality reduction, the computer power required to process the data is reduced. Furthermore, it is beneficial for extracting dominating characteristics such as rotational and positional invariant components, allowing the model to be efficiently trained.
- *FC layers (Fully Connected layers)*: they have two primary tasks, i.e., i) they link all the neurons of the preceding layers to build the numerous identification classes exhibited in the previous layers based on a given probability, and ii) they perform the final classification. They are fed with an input volume (whatever the output of the

¹ Plasma value of cardiac troponins (cTn) which regulate the contraction of the heart.

previous convolutional layers, ReLU, or pool layers was) and create a vector of size N , mirroring the number of classes from which the network must choose.

There are different types of CNNs based on their dimensions, such as 1-, 2-, or 3- dimension CNN. Among them, the differences are in the structure of the input data and in the convolution kernel, which is the filter that moves across them. In this work, we use CNN with two dimensions, namely CNN-2D.

4. Approach

The approach proposed in this work aims to perform the classification of cardiac diseases in patients divided into four groups: i) healthy ones, ii) those affected by myocardial pathologies, iii) those with a history of myocardial infarction, and iv) those affected by arrhythmia. Figure 2 shows the proposed approach in detail.

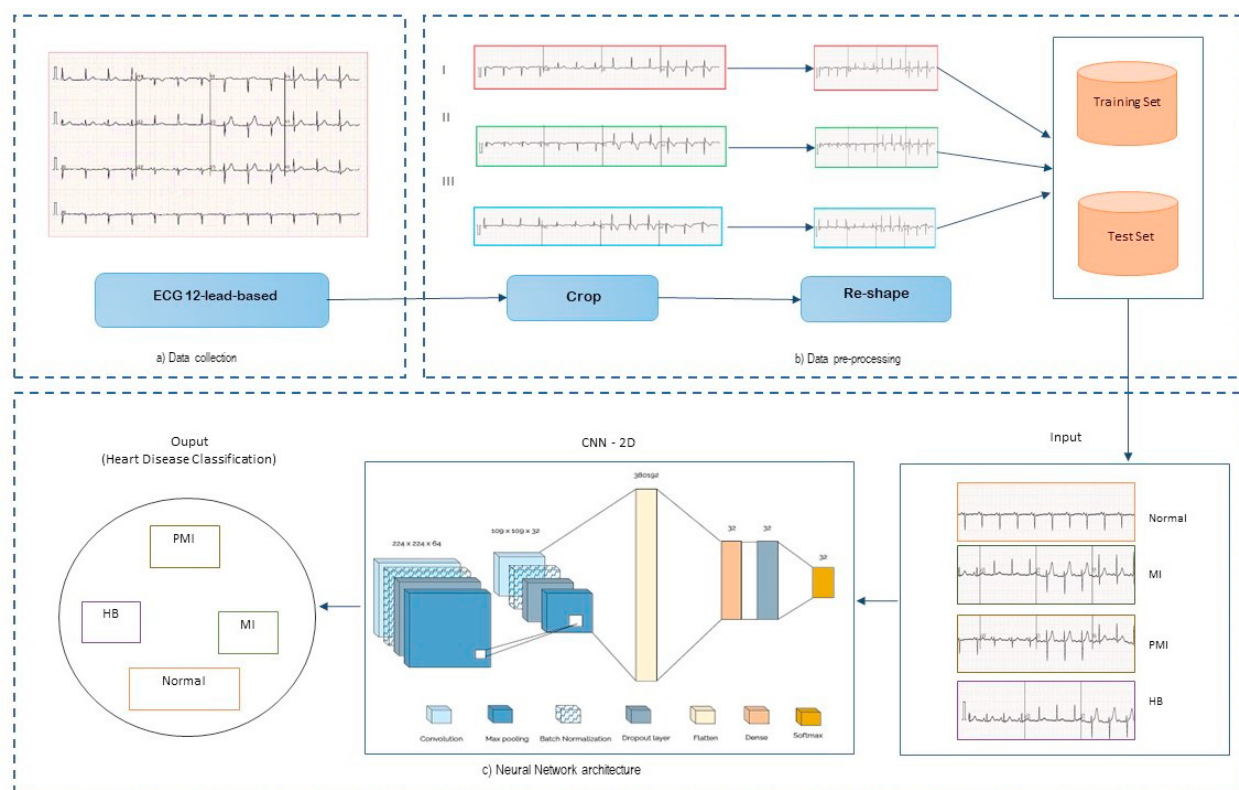


Fig. 2. Experimental approach and considered deep neural network architecture.

In particular, Subfigure 2(a) shows the ECG collection phase that uses the data from the known dataset [26], where the ECG images of healthy and ill patients are collected from the 12-lead-based sensors and divided into three different groups.

The next phase, shown in Figure 2(b), concerns the generation of three new datasets, one for each level of the ECG, through two pre-processing steps.

The first one is performed on the ECG image (in .jpg format) and concerns the electrode segmentation. Indeed, an ECG records the result of the interaction and summation of the electrical activity of each individual cell of the myocardial muscle fibre and its conduction system. This is why it is made up of waves and intervals.

From the images of the proposed ECGs, it is evident that the latter can be read through three different tracings. To understand if there is a trace that can better grasp the difference between the three pathologies, we decided to crop each ECG image in three separate bands, namely 'BAND I', 'BAND II', and 'BAND III'; each one referring to an

area of the ECG itself. Specifically, 'BAND I' includes the 'I', 'aVR', 'V1', and 'V4' electrodes; 'BAND II' includes the 'II', 'aVL', 'V2', and 'V5' electrodes; while 'BAND III' includes the 'III', 'aVF', 'V3', and 'V6' electrodes.

The second step was cropping, i.e., the bands were reshaped for all the images because they had a big dimension in terms of pixels and this took too much time for training and testing the neural network models.

Finally, after these two steps of the pre-processing phase, the obtained datasets were divided into train, test, and validation sets with a split ratio of 60/20/20 respectively.

Subfigure 2(c) reports the architecture of the considered neural network model. Networks are built starting from an input layer, in our case study made up of the images of a specific band of the ECG image for the 3 classes of cardiac pathologies and the fourth class of healthy control patients.

The input layer is followed by more layers, with different specific aims, which are detailed in Subsection 4.2. The last layer, i.e., the output layer, returns the result of the classification of the instances, in our case study the output is the classification of the heart diseases for each band of the ECG. However, before carrying out the multiclassification on the different bands created after the pre-processing of the images, binary and multi-classification of the ECGs are carried out on the full images.

4.1. Data description

The data used for this work are publicly available [26] and are composed of ECG images of cardiac patients coming from the Ch. Pervaiz Elahi Institute of Cardiology Multan (Pakistan), which aims to help the scientific community for researching cardiovascular diseases.

The dataset contains ECG images of 928 patients. All the ECG images (12-lead-based) were recorded with the Telehealth ECG diagnostic tool and they are larger than 800KB.

They have been divided into four different groups of images referring to the different heart pathologies:

- *Myocardial infarction patients (MI)*: this term refers to the necrosis of a part of the heart muscle following the obstruction of one or more coronary arteries, i.e. the arteries responsible for supplying blood to the heart. This group is made of 239 ECG images.
- *Patients that have abnormal heartbeat (HB)*: this pathology includes patients with arrhythmias, occurring when the heart beats too slowly, too fast, or irregularly. Usually, the heart beats with a frequency that varies in adults from 60 to 100 beats per minute with a regular cadence. We have 233 ECG images for this disease.
- *Patients that have had a previous MI (PMI)*: 172 ECG images of patients with a previous history of MI were put in this group.
- *Healthy people (Normal)*: in this group of images, the ECGs of 284 subjects with no cardiac pathologies at all were gathered; they are used as a control set.

4.2. CNN-2D settings and validation metrics

To classify the ECG images we used a CNN-2D architecture described in detail in subsection 3.2. In particular, Table 1 summarizes the model configuration of the proposed CNN-2D.

The first column of the Table indicates the level of the layer, the second column describes the type of layer used in the reference level, the third column shows the shape of the output obtained for that level, and the last column indicates the size of the kernel where it is used.

Other specific settings of the used CNN-2D network model are the following:

- *Activation function*: we have applied the rectified linear unit activation function via '*relu activation*';
- *Optimizer*: in our analyses, stochastic gradient descent (SGD), which is often used to train deep learning models, has been replaced by *Adam* because it needs shorter processing times and a smaller set of tuning parameters [27];
- *Dropout rate*: it is usually used to improve the generalization of the network because it reduces the over-fitting of neural networks. In our study, we set the dropout rate equal to 0.15;
- *Batch size*: it is a hyperparameter that defines the number of samples to process before updating the parameters of the internal model. We set it equal to 32;

Table 1. Configuration of the used CNN-2D.

LAYER	TYPE	OUTPUT SHAPE	KERNEL SIZE
I	Convolution	(222, 222, 64)	3
II	Batch Normalization	(222, 222, 64)	-
III	Drop Out	(222, 222, 64)	-
IV	Pooling	(111, 111, 64)	2
V	Convolution	(109, 109, 32)	3
VI	Batch Normalization	(109, 109, 32)	-
VII	Drop Out	(109, 109, 32)	-
VIII	Flatten	(380192)	-
IX	Dense	32	-
X	Drop Out	32	-
XI	Softmax	4	-

- *Epochs*: it defines the number of times the learning algorithm will run on the entire training dataset. We trained the proposed network architecture with 100 epochs.

Tensorflow², Keras³, and the Python programming language were used to implement the network's architectural designs. TensorFlow is an open-source software framework for artificial intelligence and machine learning that may be used for various applications, and it is particularly focused on deep neural networks training and testing [28]. A Python API for artificial neural networks is provided by the open-source software package known as Keras, which also provides a user-friendly Tensorflow library interface.

In order to validate the suggested models, the confusion matrix is used on the basis of the results of the classification; consequently, we measured the following validation metrics: true positives (TP) and the true negatives (TN), i.e., the correctly classified cases; false positives (FP) and false negatives (FN), i.e., the misidentified cases. These quantities are fundamental for the construction of the following metrics: *Accuracy*, *Precision*, *Recall*, *F1-Score*. In addition, we calculated the loss function; in particular, we used the *sparse categorical cross-entropy* as a probabilistic loss function. This type of loss function is used when the target has three or more classes and has the benefit of using a single integer rather than a whole vector for a class, saving memory and calculation time. Finally, during training, the accuracy and loss functions have been examined for each epoch and for all the considered network configurations.

5. Results

This section contains a description and explanation of the results we obtained in our experiments. In particular, we discuss our findings and compare the results of our approach, based on ECG images, with the results obtained from simple image classification without applying any pre-processing.

In Table 2 we report the results of the full image classification. In particular, the first part of the table reports the results of the binary classification, while the second part of the table shows the results of the multi-classification. The overall accuracy of the CNN-2D on the full images is 91% for binary classification, wherein ill patients are the best-classified ones, and 80% for the recognition of the different heart diseases.

The results show that the analyses performed on ECG images are good on both classifications; in particular, the multiclassification analysis displays that myocardial infarction patients are well recognized from the ECG images, with a precision equal to 100%.

For a better comprehension of the results, in Table 3, we report the confusion matrices of the binary classification in light blue and of the multiclassification in yellow. Most of the patients are correctly classified in the binary analysis, in fact, the main diagonal of the matrix reports higher results than the other cells. The same does not happen in the multiclass analysis, wherein patients with a history of myocardial infarction (PMI) are the worst classified. This second finding reveals that ECG images tend to classify PMI patients as HB patients; however, this result is not incorrect at

² <https://www.tensorflow.org/>

³ <https://keras.io/>

Table 2. Results on full ECG images.

Binary classification				Multiclassification			
Patient type	Precision	Recall	F1-score	Patient type	Precision	Recall	F1-score
Healthy patients	0.89	0.86	0.87	HB	0.78	0.87	0.82
Ill patients	0.93	0.94	0.94	MI	1.00	0.98	0.99
				PMI	0.52	0.38	0.44

all, in fact, most PMI patients, given the previous myocardial infarction, had an abnormal heartbeat during their life and for this reason, the ECG between a PMI patient and an HB patient can be confused.

Table 3. Confusion matrices of binary and multiclassification on full ECG images.

Binary classification			Multiclassification			
Observed	Predicted		Observed	Predicted		
	Healthy patients	Ill patients		HB	MI	PMI
Healthy patients	54	9	HB	87	0	13
Ill patients	7	116	MI	0	46	1
			PMI	24	0	15

Table 4 reports the values of the considered validation metrics and the overall accuracy of classification when considering separately the different bands of an ECG. The first column, in particular, indicates the class of patient, the second, third and fourth columns show the metrics for band I, whilst the columns from the fifth to the seventh report the metrics of band II, and the remaining columns the results of band III. Furthermore, in the last row, it is reported the overall accuracy for each band. The comparison across the accuracy of the different bands points out that the second and third ECG bands are more accurate for the classification than the first one.

As described in the previous sections, we performed band analysis to evaluate the possibility of catching some characterization in the trace between the different ECGs. In Table 4, we reported the results of band analysis by comparing all the types of patients. The comparison between the multiclassification of the entire ECGs (Table 2) and Table 4 highlights some increases in the overall accuracy values. In particular, the second and third bands demonstrate an improvement in the overall accuracy of about 7%; only the first band reports a decrease in this metric.

Table 4. Results of the CNN-2D for all types of patients considering the band-based analysis.

PATIENT	BAND I			BAND II			BAND III		
	PRECISION	RECALL	F1-SCORE	PRECISION	RECALL	F1-SCORE	PRECISION	RECALL	F1-SCORE
HEALTHY	0.89	0.87	0.88	0.92	0.94	0.93	0.91	0.98	0.95
HB	0.83	0.81	0.82	0.60	0.86	0.70	0.86	0.86	0.85
MI	0.59	0.56	0.75	0.99	0.82	0.84	0.99	0.94	0.97
PMI	0.33	0.03	0.06	0.96	0.96	0.85	0.63	0.88	0.73
TOTAL ACCURACY	0.74			0.87			0.87		

The improvement in the multiclassification performance by using band analysis is more appreciable if we look at the results by considering the evaluation metrics for each class of patients. In particular, the results in bold in Table 4 show how different heart diseases are classified differently by different bands of the ECG image itself: the first band of the ECG better classifies healthy patients and subjects affected by abnormal heartbeat (HB). The second band can better discriminate the differences between healthy patients and those affected by current or past myocardial diseases (MI and PMI). Furthermore, the third band of the ECG is able to discern better healthy patients from those affected by arrhythmias and those with myocardial necrosis.

These results are very interesting from a clinical point of view, and support even more the validity of our approach exclusively based on ECG images.

6. Threats to the validity

This section examines the threats to validity that can affect this work, with particular attention to construction, internal, and external limitations.

As for the construct validity, entailing how well a set of indicators represent or reflects a concept that is not directly measurable, we can state that our approach using three different ECG image bands to classify different heart diseases is novel but tentative, and surely needs further studies to confirm its overall scientific and medical validity.

As for internal validity, it could be affected by the images used for the analyses, which may be inaccurate because they belong to real ECG recording sessions. In order to reduce this risk, we manually inspected each image to make sure that none of them had resolution problems or presented markers that could invalidate the classification. Moreover, also the size of the images could be an issue, thus a careful reshaping operation was carried out.

Finally, as regards external validity, usually concerning the generalization of the analysis, we can state that due to the way the dataset was created, there could be some mistakes and omissions. In order to mitigate this risk, we used real data from cardiac patients, but only from one dataset, even if quite large compared to other datasets used in similar studies. We are conscious, however, that other research steps in the future may confirm or confute the findings of this study in applying our research to different datasets.

7. Conclusions & Future Work

In this work, we have performed a binary classification of healthy patients and subjects suffering from a certain type of heart disease and a multiclassification analysis for the recognition of three different types of heart diseases. For these tasks, we have used for the first time, as far as we know, a novel approach exploiting only the ECG images that are fed into a CNN-2D deep neural network. Our analyses have shown some very promising results and findings.

First of all, the analysis of the entire ECG images highlights that they are able to classify not only the difference in ECGs between healthy people and ill patients, but also to classify different heart diseases. Secondly, the results based on the analysis of the different bands highlight that it is possible to characterize different cardiac diseases through the study of different parts of the ECG image. Our approach also shows that ECG images are a really effective support for the efficient classification of heart diseases without the use of the electrical signals themselves. Finally, the analysis of the different bands also demonstrates which are the best electrodes to be used to predict the different cardiac pathologies.

As for future work, we aim to explore other neural network architectures, study further ensemble architectures made up of different deep neural network classifiers, as well as to enlarge and generalize the dataset to be used for training the considered models. Moreover, we plan to compare our results to the ones obtained by other AI-based techniques applied directly to the ECG bio-electrical signals.

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